Principle of Superposition ①

- 2 travelling waves on a stretched string.

$$y_1(x,t) = y_m \sin(\overbrace{kx-\omega t}^{\alpha})$$

$$y_2(x,t) = y_m \sin(\underbrace{kx-\omega t+\phi}_{\beta})$$

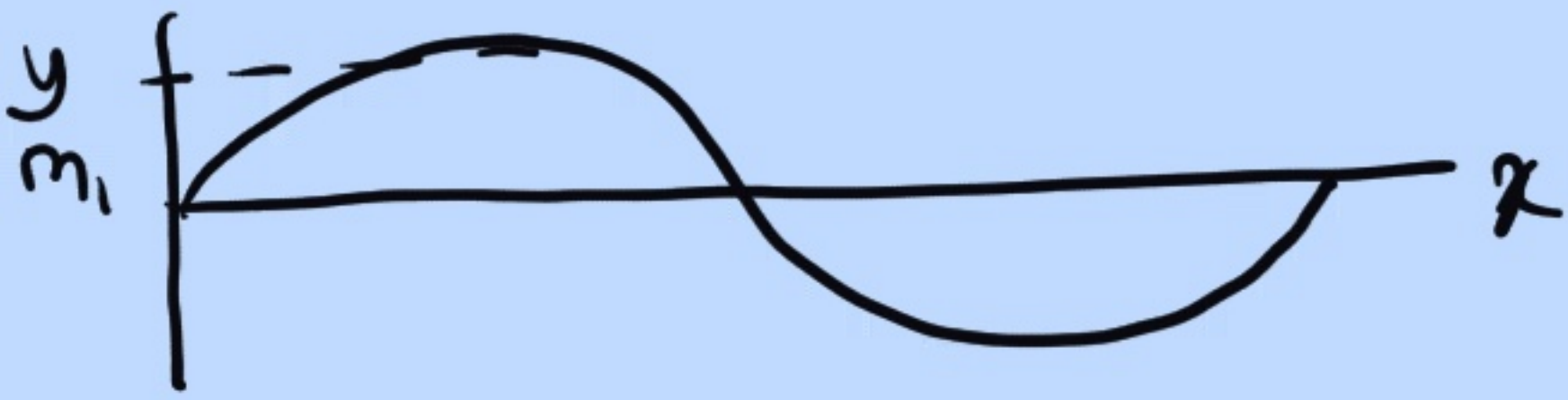
$$y_R = y_1 + y_2$$

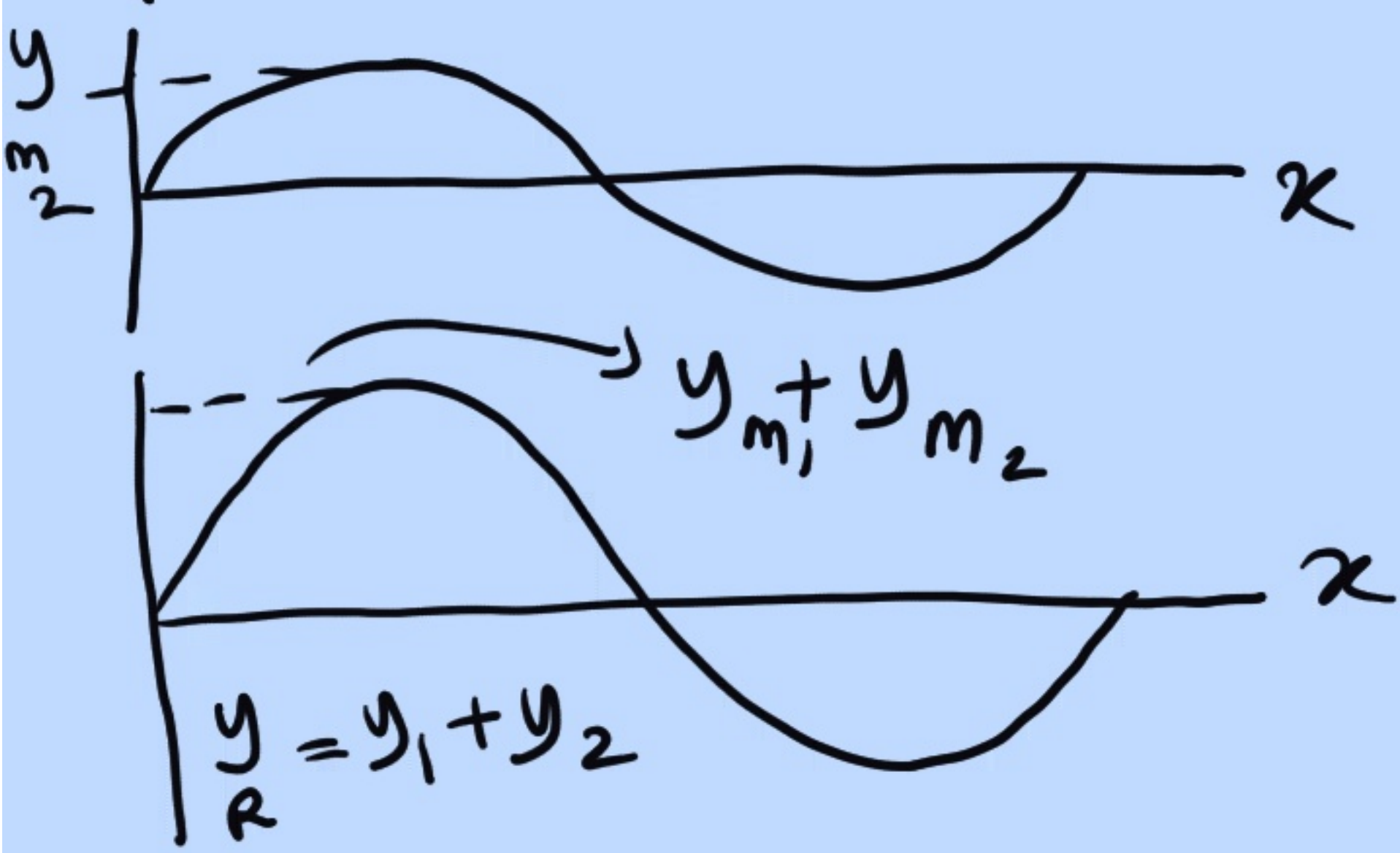
$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$y_R = \underbrace{2y_m \cos\left(\frac{\phi}{2}\right)}_{A_m} \sin\left(kx-\omega t+\frac{\phi}{2}\right)$$

Principle of Superposition

$$y_1 = y_{m_1} \sin(kx - \omega t)$$
$$y_2 = y_{m_2} \sin(kx - \omega t)$$
$$\Big\} \phi = 0$$

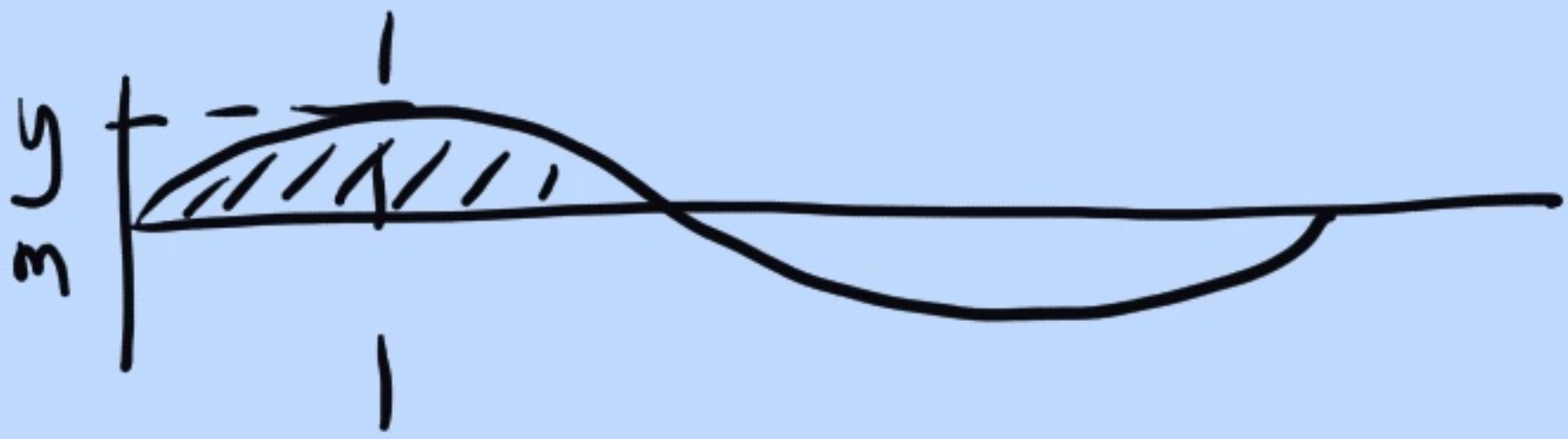


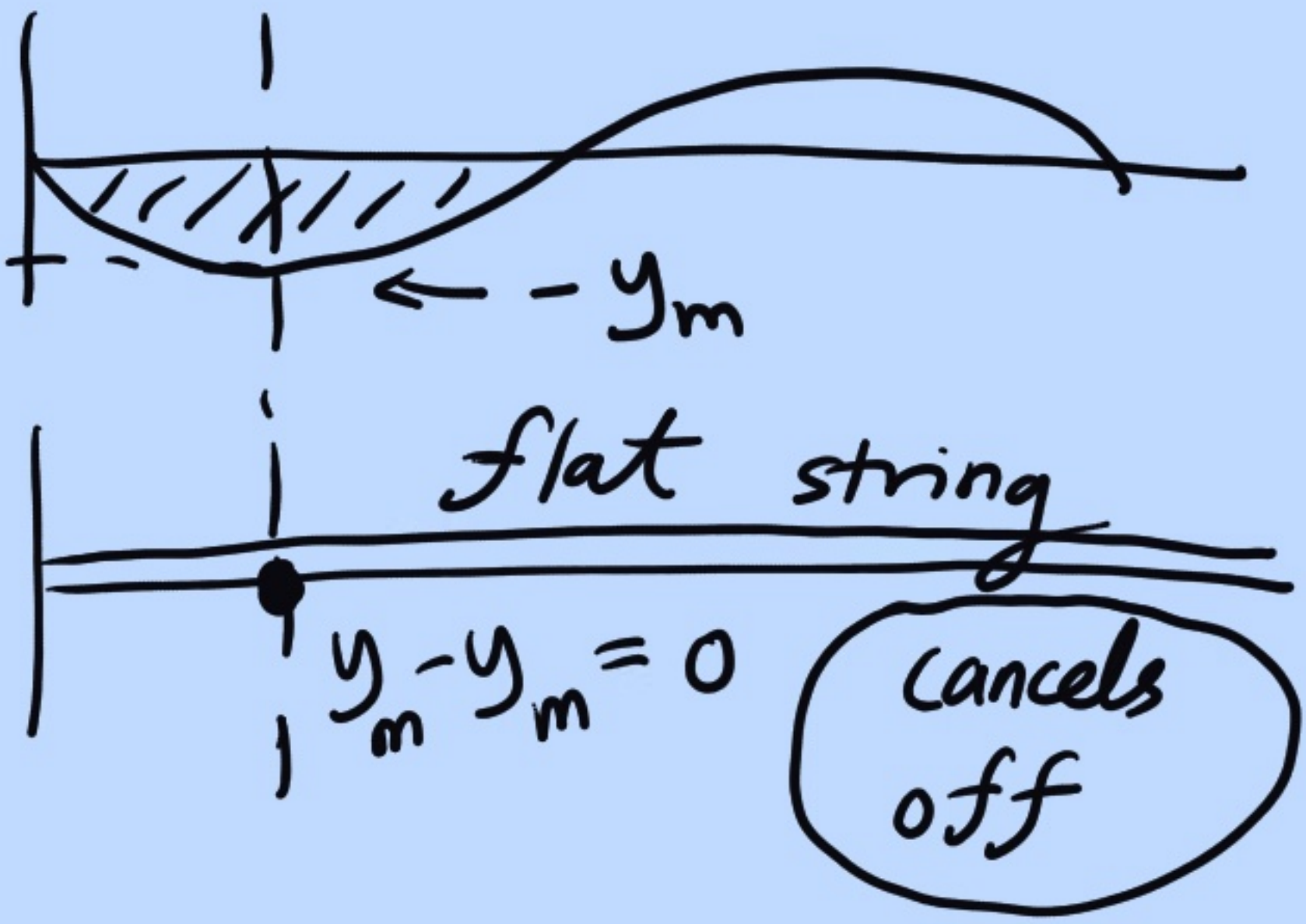


Principle of Superposition

$$y_1 = y_m \sin(kx - \omega t)$$

$$y_2 = y_m \cdot \sin(kx - \omega t + \pi) \quad \phi = \pi$$





Exercise 1: Superposition of waves

- 2 sinusodial waves are travelling with $A_1 = A_2 = 9$ mm. (on a string). $A_R = 4$ mm.
- Same λ & ω $\rightarrow$ $\lambda_1 = \lambda_2 = 0.4$ m
- Resultant wave travels in the negative x-direction by 56 cm in 8 ms & $\lambda_R = \lambda_1 = \lambda_2$.
- If $y_1 = y_m \sin(kx \pm \omega t + \phi_1)$, $\phi_1 = 0$. (Choose correct sign).

Calculate: y_m, k, ω, ϕ_R and sign for ω for [redacted] wave R.

Solution: Ex (1) (i)

a) $y_1 = y_m \cdot \sin(kx \pm \omega t)$

$y_2 = y_m \sin(kx \pm \omega t + \phi)$

$A_1 = A_2 = y_m = \underline{9 \times 10^{-3}\ m}$

b) $y_R \rightarrow \lambda_1 = \lambda_2 = \lambda_R = 0.4m$

$k_2 = \frac{2\pi}{\lambda_2} = \underline{16\ rad/m} \leftarrow k_R$

c) $V_R = \frac{\Delta x}{\Delta t} = \frac{0.56\ m}{8 \times 10^{-3}\ s} = \underline{70\ m/s}$

$V_R = \frac{\omega_R}{k_R} \rightarrow \omega_2 = \omega_R = \underline{1120\ rad/s}$

Solution : Ex(1)

d) $y_R = \underbrace{2y_m \cos\left(\frac{\phi}{2}\right)}_{A_R} \sin\left(k_R x + \omega_R t + \frac{\phi}{2}\right)$

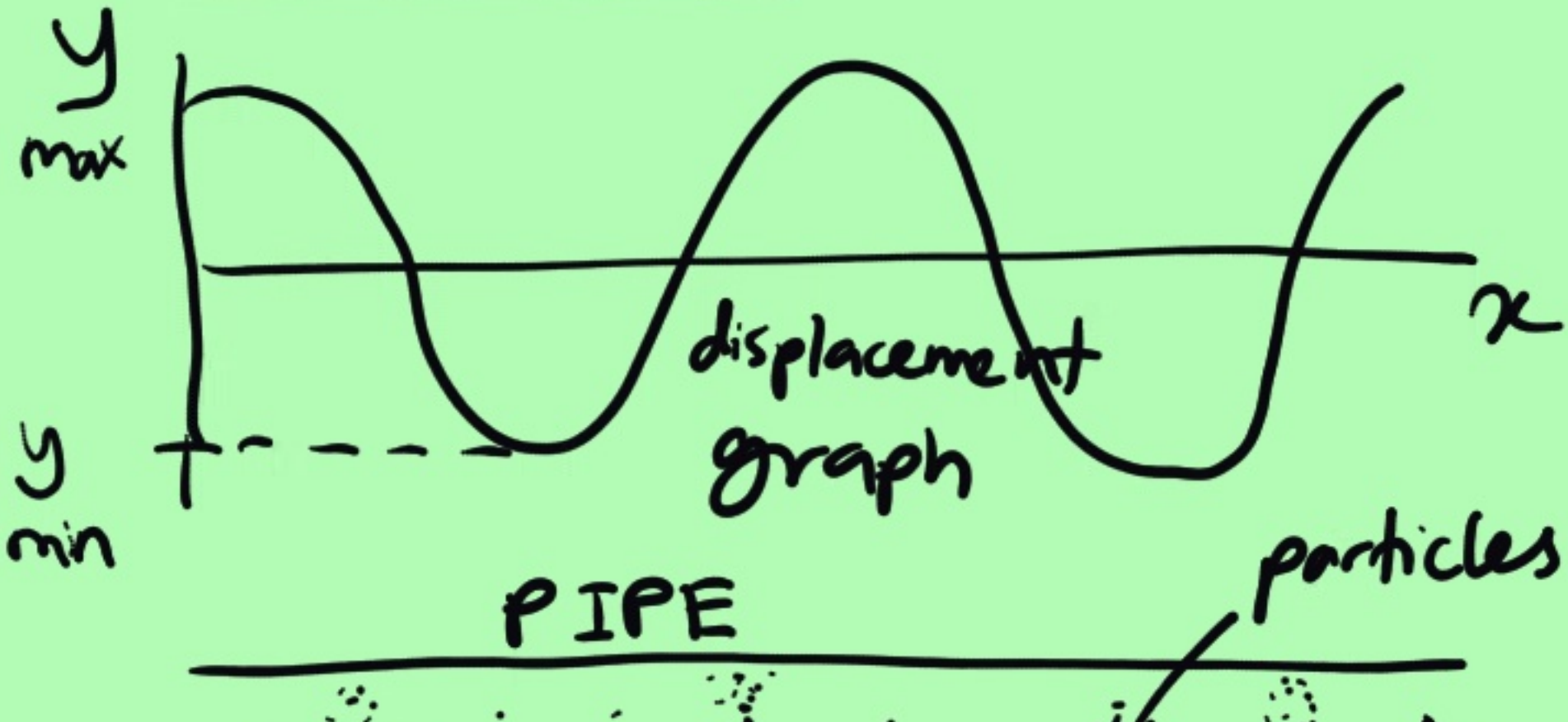
$2(9\,mm)\cos\left(\frac{\phi}{2}\right)$

$= 4\ mm$

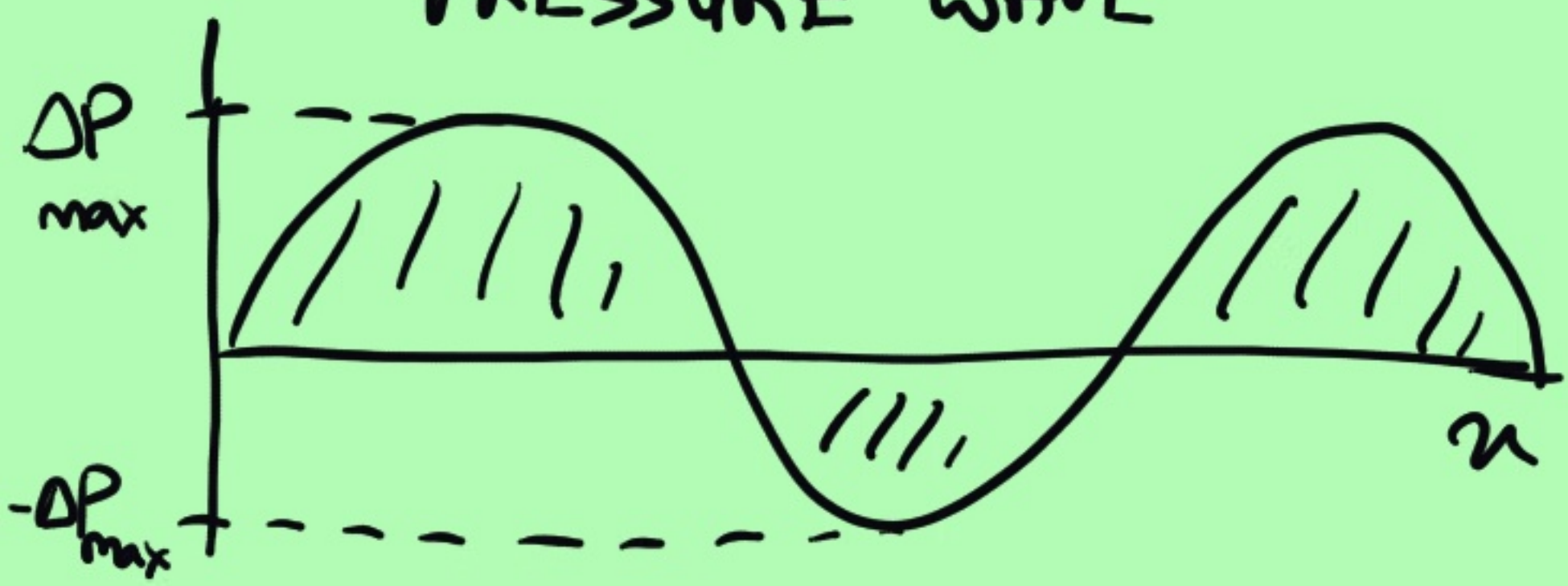
resultant wave travels in the (-) x-direction

$\phi = \underline{2.69\ rad}$.

Travelling Sound Wave



PRESSURE WAVE



Travelling Sound Wave ②

$$s(x,t) = s_m \cdot \cos(kx - \omega t)$$

↳ longitudinal displacement

$$\Delta P(x,t) = \Delta P_m \cdot \sin(kx - \omega t)$$

↳ pressure change

Amplitude Perspective

$$\Delta P_m = (v \rho \omega) s_m$$

wave speed

fluid density

unit check!

$$\Delta P_m = \frac{dP}{dx}$$

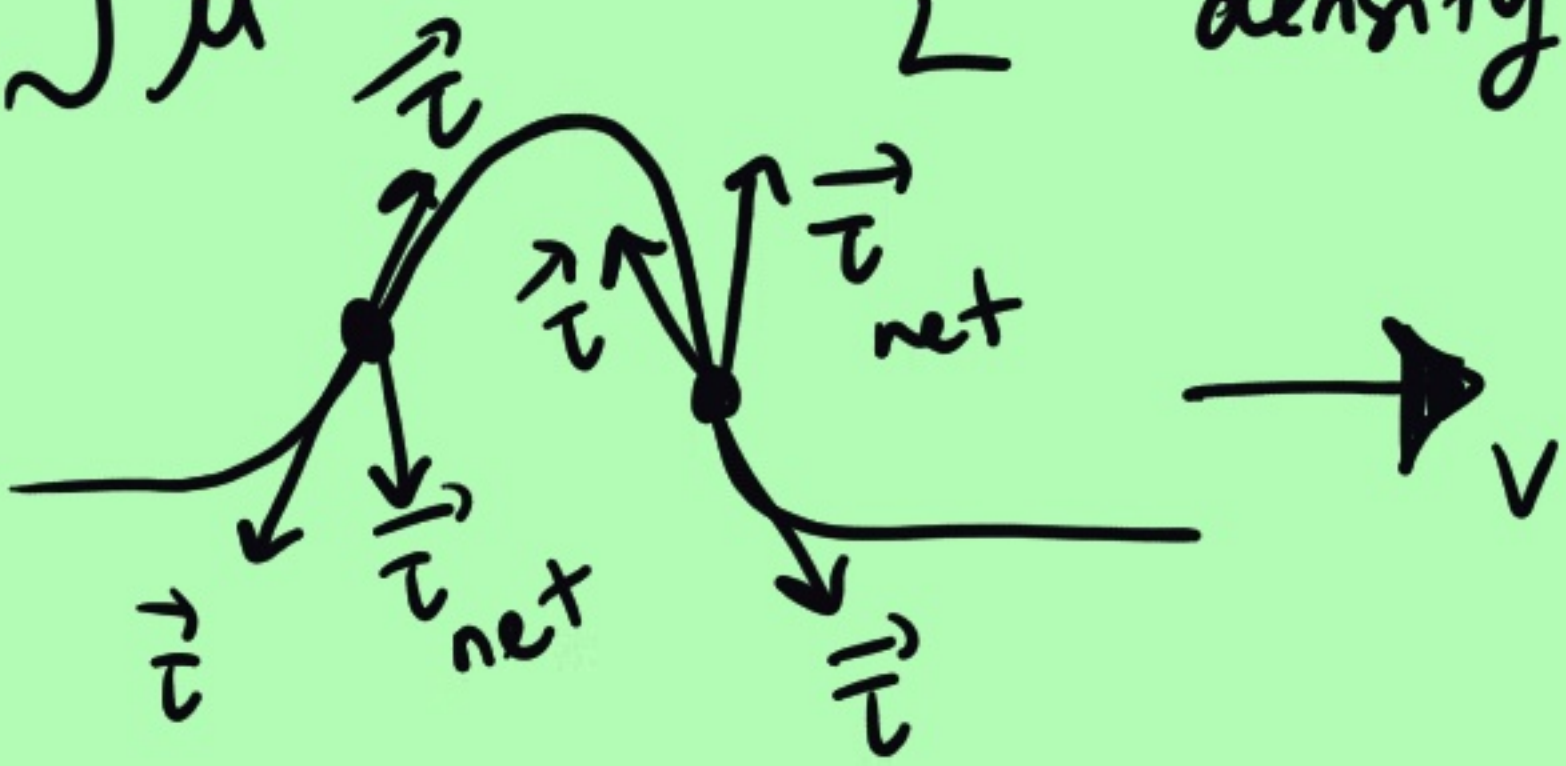
$$= \frac{F/A}{x}$$

$$= \frac{m \cdot \frac{d^2x}{dt^2}}{x A}$$

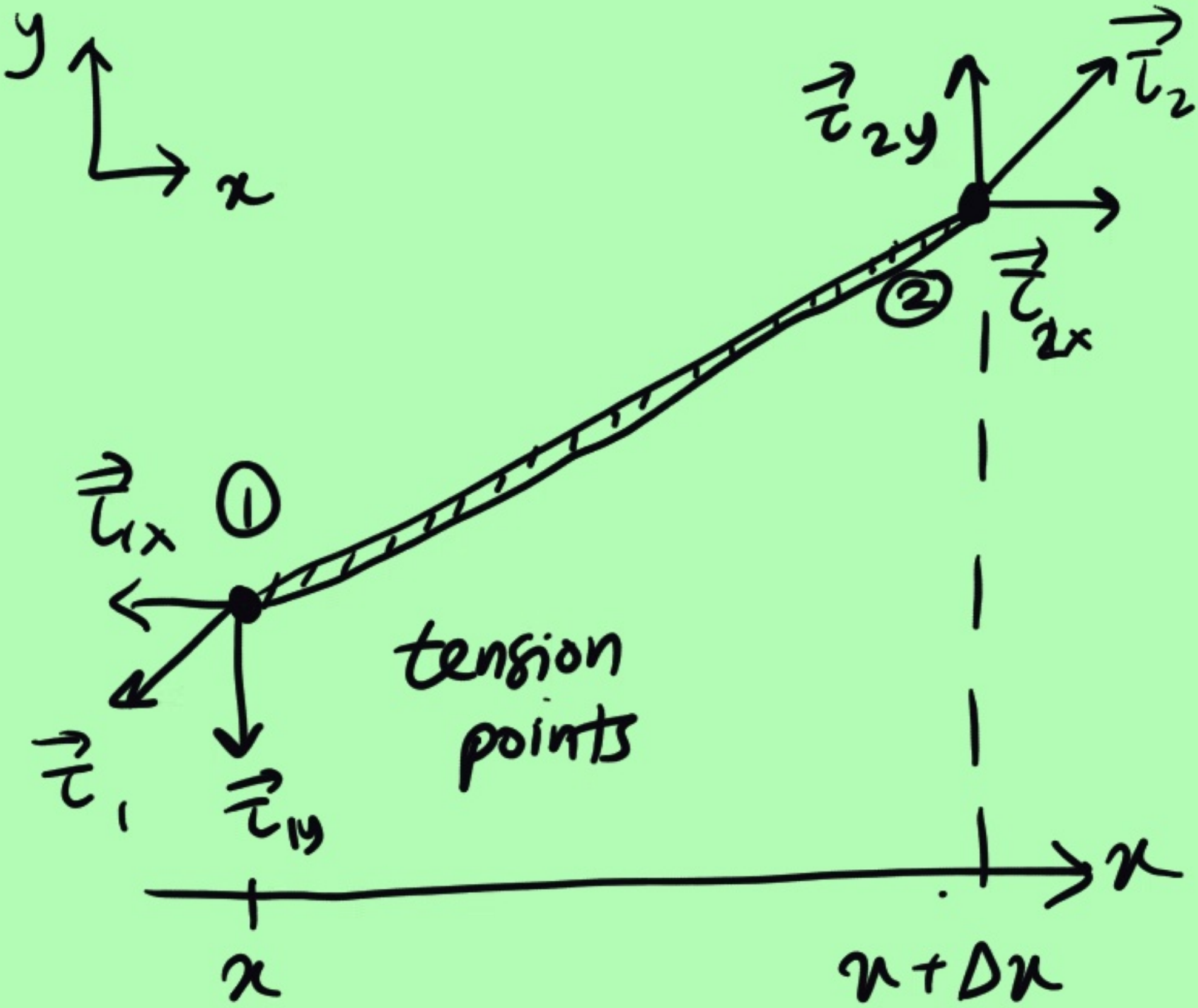
Wave Speed on a Stretched String. ③

- string with ↑ linear density
  = ↑ inertia = ↓ response time
  (slower wave speed ⇒ v)
- ↑ tension = ↑ wave speed

$$v = \sqrt{\frac{\tau}{\mu}} \qquad \mu = \frac{m}{L}$$ linear density



Wave Equation





x - component : (5)

$\tau_{1x} = \tau_{2x} = \bar{\tau}$

y - component :

$\tau_y = \tau_{2y} - \tau_{1y}$ ——— (1)

At x : $\left(\frac{\partial y}{\partial x}\right)_x = \frac{\tau_{1y}}{\tau_{1x}} = \frac{\tau_{1y}}{\bar{\tau}}$ — (2)

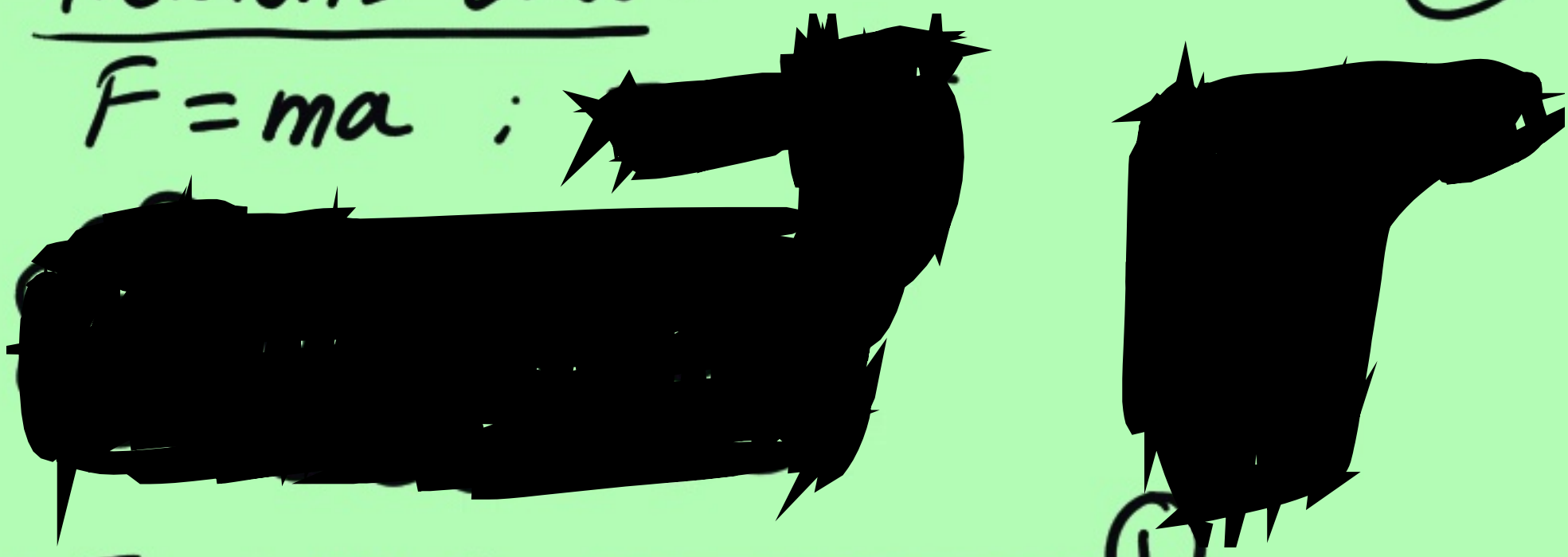
At $x+\Delta x$: $\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} = \frac{\tau_{2y}}{\tau_{2x}} = \frac{\tau_{2y}}{\bar{\tau}}$ (3)

(2) & (3) into (1)

$$\bar{\tau}_y = \bar{\tau}\left(\frac{\partial y}{\partial x}\right)_{x+\Delta x} - \bar{\tau}\left(\frac{\partial y}{\partial x}\right)_x$$

Newton's Law :

$F = ma$;



$\therefore \bar{\tau}_y = m \cdot a_y$ ——— (1)

$\mu = m / \Delta x$ — (2) Linear mass density

$$= \frac{\left[\frac{\partial y}{\partial x}\right]_{x+\Delta x} - \left[\frac{\partial y}{\partial x}\right]_{x}}{\Delta x}$$

string segment that approx. a straight line.

$$= \frac{\mu}{\tau} \cdot a_y \rightarrow \therefore \frac{\partial^2 y}{\partial x^2} = \frac{\mu}{\tau} \frac{\partial^2 y}{\partial t^2}$$

$$v = \sqrt{\tau / \mu}$$